Exercise Sheet 4: Numerical Differentiation and Integration

H By Hand; **C** Computer; **T** Theory; **E** Extra; **A** Advanced.

Recommended: H1.b; H2.a; C3; T4; C6.a; H7.a; H8.a; H10.a; C12.b.

Differentition

H1. Use the most accurate three-point formula to determine $f'(x)$ and $f''(x)$ at each data point in the following tables:

a.

x	$f(x)$
0.9	7.91
1.0	9.91
1.1	12.35
1.2	15.33

b.

x	$f(x)$
4.3	−5.71
4.5	−6.33
4.7	−6.96
4.9	−7.59

c.

x	$f(x)$
2.5	3.939
2.75	3.351
3.0	2.608
3.25	1.708
3.5	1.216

Answer

a. Use forward-difference at $x = 0.9$, centred-difference at $x = 1.0, 1.1$ and backward-difference at $x = 1.2$ with $h = 0.1$:

$$\begin{aligned}
f'(0.9) &\approx \frac{-3f(0.9) + 4f(1.0) - f(1.1)}{2 \times 0.1} = \frac{-3 \times 7.91 + 4 \times 9.91 - 12.35}{2 \times 0.1} = \frac{3.56}{0.2} = 17.8;\\
f'(1.0) &\approx \frac{f(1.1) - f(0.9)}{2 \times 0.1} = \frac{12.35 - 7.91}{2 \times 0.1} = \frac{4.44}{0.2} = 22.2;\\
f'(1.1) &\approx \frac{f(1.2) - f(1.0)}{2 \times 0.1} = \frac{15.33 - 9.91}{2 \times 0.1} = \frac{5.42}{0.2} = 27.1;\\
f'(1.2) &\approx \frac{f(1.0) - 4f(1.1) + 3f(1.2)}{2 \times 0.1} = \frac{9.91 - 4 \times 12.35 + 3 \times 15.33}{2 \times 0.1} = \frac{6.50}{0.2} = 32.5.
\end{aligned}$$

b. $f'(4.3) \approx -3.075 = -3.1\,(1\text{dp})$, $f'(4.5) \approx -3.125 = -3.1$, $f'(4.7) \approx -3.150$, $f'(4.9) \approx -3.15$.

c. $f'(2.5) \approx -2.042 = -2.04\,(2\text{dp})$, $f'(2.75) \approx -2.662$, $f'(3.0) \approx -3.286$, $f'(3.25) \approx -2.784$, $f'(3.5) \approx -1.152$.

H2. Use the five-point differentiation formulae to determine, as accurately as possible, approximations for $f'(x)$ at each data point in the following tables:

a.

x	$f(x)$
3.0	0.2902
3.25	0.6262
3.5	0.8808
3.75	1.0840
4.0	1.2530

b.

x	$f(x)$
−2.0	8.1333
−1.8	6.9987
−1.6	5.9458
−1.4	4.9748
−1.2	4.0858
−1.0	3.2789

Answer

b. Use step-size $h = 0.25$.
$f'(3.0) \approx 1.55960 = 1.560\,(3\text{dp})$, $f'(3.25) \approx 1.15693$, $f'(3.5) \approx 0.89987$, $f'(3.75) \approx 0.73720$, $f'(4.0) \approx 0.61773$.

c. Use forward-difference at $x = -2.0$, assymetrical forward-difference at $x = -1.8$, centred-difference at $x = -1.6, -1.4$ with $h = 0.2$. Use (assymetrical) forward-difference at $x = -1.2$,

$x = -1.0$ with $h = -0.2$.

$$\begin{aligned}
f'(-2.0) &\approx \frac{-25f(-2.0) + 48f(-1.8) - 36f(-1.6) + 16f(-1.4) - 3f(-1.2)}{12 \times 0.2} \\
&= \frac{-25 \times 8.1333 + 48 \times 6.9987 - 36 \times 5.9458 + 16 \times 4.9748 - 3 \times 4.0858}{12 \times 0.2} \\
&= \frac{-203.3325 + 335.9376 - 214.0488 + 79.5968 - 12.2574}{2.4} = \frac{-14.1043}{2.4} \\
&= -5.8768; \\
f'(-1.8) &\approx \frac{-3f(-2.0) - 10f(-1.8) + 18f(-1.6) - 6f(-1.4) + f(-1.2)}{12 \times 0.2} \\
&= \frac{-3 \times 8.1333 - 10 \times 6.9987 + 18 \times 5.9458 - 6 \times 4.9748 + 4.0858}{12 \times 0.2} \\
&= \frac{-24.3999 - 69.9870 + 107.0244 - 29.8488 + 4.0858}{2.4} = \frac{-13.1255}{2.4} \\
&= -5.4690; \\
f'(-1.6) &\approx \frac{f(-2.0) - 8f(-1.8) + 8f(-1.4) - f(-1.2)}{12 \times 0.2} \\
&= \frac{8.1333 - 8 \times 6.9987 + 8 \times 4.9748 - 4.0858}{12 \times 0.2} \\
&= \frac{8.1333 - 55.9896 + 39.7984 - 4.0858}{2.4} = \frac{-12.1437}{2.4} \\
&= -5.0599; \\
f'(-1.4) &\approx \frac{f(-1.8) - 8f(-1.6) + 8f(-1.2) - f(-1.0)}{12 \times 0.2} \\
&= \frac{6.9987 - 8 \times 5.9458 + 8 \times 4.0858 - 3.2789}{12 \times 0.2} = \frac{-11.1602}{2.4} \\
&= -4.6501; \\
f'(-1.2) &\approx \frac{-f(-1.8) + 6f(-1.6) - 18f(-1.4) + 10f(-1.2) + 3f(-1.0)}{12 \times 0.2} \\
&= \frac{-6.9987 + 6 \times 5.9458 - 18 \times 4.9748 + 10 \times 4.0858 + 3 \times 3.2789}{12 \times 0.2} = \frac{-10.1756}{2.4} \\
&= -4.2398; \\
f'(-1.0) &\approx \frac{3f(-1.8) - 16f(-1.6) + 36f(-1.4) - 48f(-1.2) + 25f(-1.0)}{12 \times 0.2} \\
&= \frac{3 \times 6.9987 - 16 \times 5.9458 + 36 \times 4.9748 - 48 \times 4.0858 + 25 \times 3.2789}{12 \times 0.2} = \frac{-9.1898}{2.4} \\
&= -3.8291.
\end{aligned}$$

C3. Estimate the first derivative of $f(x) = 1/(3 - 2x^2)$ at $x = 1$ using (i) the three-point centred-difference formula, the (ii) three-point forward difference formula, and (iii) the five-point centred difference formula, and the second derivative at $x = 1$ using (iv) the three-point fomulae and (v) the five-point centred formula. Use $h = 10^{-n}$ for $n = 0, 1, \ldots, 12$. Compute the errors in the approximations and verify the order of each method.

Answer Exact values $f(1) = 1$, $f'(1) = 4$ and $f''(1) = 36$.
With $h = 0.1$, find (i) $f'(1) \approx 4.998$, (ii) $f'(1) \approx -22.184$, (iii) $f'(1) \approx 0.203$, (iv) $f''(1) \approx 44.878$, (v) $f''(1) \approx 2.214$.
With $h = 0.01$, find (i) $f'(1) \approx 4.0080$, (ii) $f'(1) \approx 3.9816$, (iii) $f'(1) \approx 3.999936$, (iv) $f''(1) \approx 36.0713$, (v) $f''(1) \approx 35.999431$.
With $h = 0.001$, find (i) $f'(1) \approx 4.000080$, (ii) $f'(1) \approx 3.999838$, (iii) $f'(1) \approx 3.9999999937$, (iv) $f''(1) \approx 36.000712$, (v) $f''(1) \approx 35.9999999429$.
Error of (i,ii,iv) is $O(h^2)$, with (ii) being twice that of (i). Error if (iii,v) is $O(h^4)$

T4. Let ϵ denote a bound on the total error made in evaluating a function f, and M denotes a bound for the third derivative of f. Show that the total error in estimating $f'(x)$ using the three-point centred-difference formula is

$$e(h) = \frac{\epsilon}{h} + \frac{h^2}{6}M.$$

Which value of h should you choose to minimise the error?

Answer Error is minimised when $h \approx \sqrt[3]{3\epsilon/M}$.

A5. Show that if f is 5-times continuously differentiable, then

$$f'(x) = \frac{-3f(x-h) - 10f(x) + 18f(x+h) - 6f(x+2h) + f(x+3h)}{12h} - \frac{h^4}{20}f^{(5)}(\xi)$$

for some ξ between $x-h$ and $x+3h$.

Show that if f is 6-times continuously differentiable, then

$$f''(x) = \frac{-f(x-2h) + 16f(x-h) - 30f(x) + 16f(x+h) - f(x+2h)}{12h^2} - \frac{h^4}{90}f^{(6)}(\xi)$$

for some ξ between $x-2h$ and $x+2h$.

Use these formula to find better estimates to $f'(x)$ for the data entries next to the endpoints in Question 2, and to estimate the second derivatives.

Note: Other five-point second-derivative formulae are

$$f''(x) = \frac{11f(x-h) - 20f(x) + 6f(x+h) + 4f(x+2h) - f(x+3h)}{12h^2} - \frac{19}{360}h^4 f^{(6)}(\xi)$$
$$f''(x) = \frac{35f(x) - 104f(x+h) + 114f(x+2h) - 56f(x+3h) + 11f(x+4h)}{12h^2} - \frac{119}{90}h^4 f^{(6)}(\xi)$$

Integration

C6. Use Matlab's built-in `integral` command to evaluate the following integrals:

a. $\displaystyle\int_0^1 e^{-x^2/2}\,dx$ **b.** $\displaystyle\int_0^3 \frac{\sin x}{x}\,dx$

Answer

a. `integral(@(x)exp(-x.^2/2),0,1)`

b. `integral(@(x)sin(x)./x,0,3)`

H7. Use (i) the midpoint rule, (ii) the trapezoidal rule and (iii) Simpson's rule with the indicated values of n, or the indicated h to approximate the following integrals.

a. $\displaystyle\int_1^3 x\log x\,dx, \quad n=4$ **b.** $\displaystyle\int_0^2 \frac{1}{x^2+9}\,dx, \quad h=\tfrac{1}{3}$

c. $\displaystyle\int_0^1 e^{-x^2/2}\,dx, \quad n=8$ **d.** $\displaystyle\int_0^3 \frac{\sin x}{x}\,dx \quad h=0.5$

Note: Take $\sin(0)/0 = \lim_{h\to 0}\sin(h)/h = 1$.

Answer

a. $h = (b-a)/n = (3-1)/4 = 0.5$.

$$M_4(f;1,3) = M(f;[1.0,1.5,2.0,2.5,3.0]) = \frac{3-1}{4}\big(f(1.25)+f(1.75)+f(2.25)+f(2.75)\big)$$
$$= 0.5 \times (0.2783 + 0.9793 + 1.8246 + 2.7819) = 0.5 \times 5.8648 = 2.9324$$

$$T_4(f;1,3) = T(f;[1.0,1.5,2.0,2.5,3.0]) = \frac{3-1}{4}\big(\tfrac{1}{2}f(1.0)+f(1.5)+f(2.0)+f(2.5)+\tfrac{1}{2}f(3.0)\big)$$
$$= 0.5 \times (\tfrac{1}{2}\times 0.0000 + 0.6082 + 1.3863 + 2.2907 + \tfrac{1}{2}\times 3.2958) = 0.5 \times 5.9333 = 2.9666$$

$$S_4(f;1,3) = S(f;[1.0,1.5,2.0,2.5,3.0]) = \frac{3-1}{4}\frac{1}{3}\big(f(1.0)+4f(1.5)+2f(2.0)+4f(2.5)+f(3.0)\big)$$
$$= 0.5 \times \tfrac{1}{3}(0.000000 + 4\times 0.608197 + 2\times 1.386294 + 4\times 2.290727 + 3.295837)$$
$$= 0.5 \times \tfrac{1}{3} \times 17.664124 = 0.5 \times 5.888041 = 2.944021 .$$

b. $n = (b-a)/h = 6$, $M_6(f;0,2) = 0.1961$, $T_6(f;0,2) = 0.1957$, $S_6(f;0,2) = 0.196001$.

c. $h = 0.125$, $M_8(f;0,1) = 0.8560$, $T_8(f;0,2) = 0.8548$, $S_8(f;0,2) = 0.855626$.

d. $n = 6$, $M_6(f;0,2) = 1.8523$, $T_6(f;0,2) = 1.8414$, $S_6(f;0,2) = 1.848704$.

H8. Compute the Romburg integral $R_{3,3}$ approximating

a. $\displaystyle\int_1^3 x\log x\,dx$ **b.** $\displaystyle\int_0^2 e^{-x^2}\,dx$

Compare your answers with those obtained previously by using Simpson's rule, and explain any correspondences between your results.

Answer

a.

$$T_1(f;1,3) = R_{0,0} = \tfrac{3-1}{1}\big(\tfrac{1}{2}f(1.0)+\tfrac{1}{2}f(3.0)\big)$$
$$= 2.0 \times (\tfrac{1}{2}\times 0.000000 + \tfrac{1}{2}\times 3.29583687) = 3.29583687$$
$$T_2(f;1,3) = R_{1,0} = \tfrac{3-1}{2}\big(\tfrac{1}{2}f(1.0)+f(2.0)+\tfrac{1}{2}f(3.0)\big)$$
$$= \tfrac{1}{2}T_1(f;1,3) + \tfrac{3-1}{2}\big(f(2.0)\big)$$
$$= \tfrac{1}{2}\times 3.29583687 + 1.0 \times 1.38629436 = 3.03421279$$
$$T_4(f;1,3) = R_{2,0} = \tfrac{3-1}{4}\big(\tfrac{1}{2}f(1.0)+f(1.5)+f(2.0)+f(2.5)+\tfrac{1}{2}f(3.0)\big)$$
$$= \tfrac{1}{2}T_2(f;1,3) + \tfrac{3-1}{4}\big(f(1.5)+f(2.5)\big)$$
$$= \tfrac{1}{2}\times 3.034213 + 0.5 \times (0.608197662162247{+}2.290726829685388) = 2.96656864$$
$$T_8(f;1,3) = R_{3,0} = \tfrac{3-1}{8}\big(\tfrac{1}{2}f(1.0)+f(1.25)+f(1.5)+f(1.75)+f(2.0)+f(2.25)+f(2.5)+f(2.75)+\tfrac{1}{2}f(3.0)\big)$$
$$= \tfrac{1}{2}T_4(f;1,3) + \tfrac{3-1}{8}\big(f(1.25)+f(1.75)+f(2.25)+f(2.75)\big)$$
$$= \tfrac{1}{2}\times 2.96656864 + 0.25 \times (0.27892944{+}0.97932763{+}1.82459299{+}2.78190251) = 2.94947246$$

$$R_{1,1} = R_{1,0} + \tfrac{1}{3}(R_{1,0} - R_{0,0}) = 3.03421279 + \tfrac{1}{3}\times(3.03421279 - 3.29583687) = 2.94700477$$
$$R_{2,1} = R_{2,0} + \tfrac{1}{3}(R_{2,0} - R_{1,0}) = 2.96656864 + \tfrac{1}{3}\times(2.96656864 - 3.03421279) = 2.94402059$$
$$R_{3,1} = R_{3,0} + \tfrac{1}{3}(R_{3,0} - R_{2,0}) = 2.94947246 + \tfrac{1}{3}\times(2.94947246 - 2.96656864) = 2.94377373$$
$$R_{2,2} = R_{2,1} + \tfrac{1}{3}(R_{2,1} - R_{1,1}) = 2.94402059 + \tfrac{1}{15}\times(2.94402059 - 2.94700477) = 2.94382165$$
$$R_{3,2} = R_{3,1} + \tfrac{1}{3}(R_{3,1} - R_{2,1}) = 2.94377373 + \tfrac{1}{15}\times(2.94377373 - 2.94402059) = 2.94375728$$
$$R_{3,3} = R_{3,2} + \tfrac{1}{3}(R_{3,2} - R_{2,2}) = 2.94375728 + \tfrac{1}{63}\times(2.94375728 - 2.94382165) = 2.94375626$$

b.

$R_{0,0} = T_1 = 1.01831564$			
$R_{1,0} = T_2 = 0.87703726$	$R_{1,1} = 0.82994447$		
$R_{2,0} = T_4 = 0.88061863$	$R_{2,1} = 0.88181243$	$R_{2,2} = 0.88527029$	
$R_{3,0} = T_8 = 0.88170379$	$R_{3,1} = 0.88206551$	$R_{3,2} = 0.88208238$	$R_{3,3} = 0.88203178$

The $R_{i,1}$ values are the same as $S_{2^i}(f;a,b)$.

H9. The following data (x,y) are arise from the function $y=f(x)$.

x	1.4	1.6	1.8	2.0	2.2	2.4	2.6
y	2.2575	2.8773	3.1409	3.0833	3.0592	3.0391	3.3111

Approximate $\int_{1.4}^{2.6} f(x)\,dx$ as accurately as possible using (i) the midpoint rule, (ii) the trapezoidal rule and (iii) Simpson's rule.

Answer (i) Can only use $n=3$ subdivisions, since need the midpoints of the intervals. For (ii) can use $n=6$ subdivisions. For (iii) use $m=3$ quadratic pieces, and all data points.

$$\begin{aligned}
M_3(f;1.4,2.6) &= M(f;[1.4,1.8,2.2,2.6]) = \tfrac{2.6-1.4}{3}\big(f(1.6)+f(2.0)+f(2.4)\big)\\
&= 0.4\times(2.8773+3.0833+3.0391)\\
&= 3.59988 = 3.5999\,(4\text{dp});\\
T_6(f;1.4,2.6) &= \tfrac{2.6-1.4}{6}\big(\tfrac{1}{2}f(1.4)+f(1.6)+f(1.8)+f(2.0)+f(2.2)+f(2.4)+\tfrac{1}{2}f(2.6)\big)\\
&= 0.2\times(\tfrac{1}{2}\times 2.2575+2.8773+3.1409+3.0833+3.0592+3.0391+\tfrac{1}{2}\times 3.3111)\\
&= 3.59682 = 3.5968\,(4\text{dp});\\
S_6(f;1.4,2.6) &= \tfrac{2.6-1.4}{6}\big(f(1.4)+4\times f(1.6)+2\times f(1.8)+4\times f(2.0)+2\times f(2.2)+4\times f(2.4)+\tfrac{1}{2}f(2.6)\big)/3\\
&= 0.2\times(2.2575+4\times 2.8773+2\times 3.1409+4\times 3.0833+2\times 3.0592+4\times 3.0391+3.3111)/3\\
&= 3.59682 = 3.5968\,(4\text{dp}).
\end{aligned}$$

H10. Use (i) the midpoint rule, (ii) the trapezoidal rule, and (iii) Simpson's rule to approximate the integrals

a. $\displaystyle\int_2^3 x^3\,dx$ **b.** $\displaystyle\int_1^3 x\log x\,dx$

Use $n=1$, $n=2$ and $n=4$ for the midpoint and trapezoidal rules, and $n=2$ and $n=4$ for Simpson's rule. Compare your results with the exact values (a) 65/4 and (b) $\frac{9}{2}\log 3-2$.

How do your results correspond to what you would expect given the order of the methods?

Answer

a.

$$\begin{aligned}
&M_1(f;2,3) = 15.625;\quad E_{M_1} = |15.625-16.25| = 0.625\\
&M_2(f;2,3) = 16.09375;\quad E_{M_2} = 0.15625;\quad E_{M_2}/E_{M_1} = 0.15625/0.625 = \tfrac{1}{4}\\
&M_4(f;2,3) = 16.2109375;\quad E_{M_4} = 0.0390625;\quad E_{M_4}/E_{M_2} = \tfrac{1}{4}\\
&T_1(f;2,3) = 17.5;\quad E_{T_1} = |17.5-16.25| = 1.25\\
&T_2(f;2,3) = 16.5625;\quad E_{T_2} = 0.3125;\quad E_{T_2}/E_{T_1} = 0.3125/1.25 = \tfrac{1}{4}\\
&T_4(f;2,3) = 16.328125;\quad E_{T_4} = 0.078125;\quad E_{T_4}/E_{T_2} = \tfrac{1}{4}\\
&S_2(f;2,3) = 16.25;\quad E_{S_2} = 0.\\
&S_4(f;2,3) = 16.25;\quad E_{S_4} = 0.
\end{aligned}$$

For midpoint and trapezoid rules, if interval width h is multiplies by $\frac{1}{2}$, error is multiplied by $\frac{1}{4}=\frac{1}{2^2}$, consistent with second-order $O(h^2)$. For Simpson's rule, method is exact for polynomial of degree 3, since error depends on $\max_{\xi\in[a,b]}|f^{(4)}(\xi)|$.

b.

$M_1(f;1,3) = 2.7726\,(4\text{dp});\quad E_{M_1} = |2.7726 - 2.9438| = 0.1712\,(4\text{dp})$
$M_2(f;1,3) = 2.8989;\quad E_{M_2} = 0.0448;\quad E_{M_2}/E_{M_1} = 0.2619\,(4\text{dp}) \approx \frac{1}{4}$
$M_4(f;1,3) = 2.9324;\quad E_{M_4} = 0.0114;\quad E_{M_4}/E_{M_2} = 0.2538 \approx \frac{1}{4}$

$T_1(f;1,3) = 3.2958\,(4\text{dp});\quad E_{T_1} = |3.2958 - 2.9438| = 0.3521\,(4\text{dp})$
$T_2(f;1,3) = 3.0342;\quad E_{T_2} = 0.0905;\quad E_{T_2}/E_{T_1} = 0.0905/0.3521 = 0.2569\,(4\text{dp}) \approx \frac{1}{4}$
$T_4(f;1,3) = 2.9666;\quad E_{T_4} = 0.0228;\quad E_{T_4}/E_{T_2} = 0.2522 \approx \frac{1}{4}$

$S_2(f;1,3) = 2.947005\,(4\text{dp});\quad E_{S_2} = |2.947005 - 2.943755| = 0.003249\,(6\text{dp}).$
$S_4(f;1,3) = 2.944021;\quad E_{S_4} = 0.000265;\quad E_{S_4}/E_{S_4} = 0.003249/0.000265 = 0.0816\,(4\text{dp}) \approx 0.0625 = \frac{1}{16}.$
$S_4(f;1,3) = 2.943774;\quad E_{S_8} = 0.000018;\quad E_{S_4}/E_{S_4} = 0.003249/0.000265 = 0.0695\,(4\text{dp}) \approx 0.0625 = \frac{1}{16}.$

For Simpson's rule, if interval width h is multiplies by $\frac{1}{2}$, error is multiplied by approximately $\frac{1}{16} = \frac{1}{2^4}$, consistent with fourth-order $O(h^4)$.

A11. Use the error formula to find a bound for the error for each integral in Question 10, and compare the bound to the actual error.

H12. For each of the data in Question 1, use your finite-difference estimates of $f'(x_i)$ and the trapezoid rule to estimate $\int_{x_0}^{x_m} \sqrt{1+f'(x)^2}\,dx$, where x_0 is the first data point and x_m the last.

C13. Write Matlab functions to approximate $\int_a^b f(x)\,dx$ using (i) the midpoint rule, (ii) the trapezoidal rule, (iii) Simpson's rule and (iv) Romburg integration with a given value of n. Test your code by computing the following integrals with the given values of n.

a. $\displaystyle\int_0^3 \frac{\sin x}{x}\,dx,\quad n = 8, 16$ **b.** $\displaystyle\int_0^1 e^{-x^2/2}\,dx,\quad n = 8, 16$ **c.** $\displaystyle\int_4^5 \frac{1}{\sqrt{x^2-9}}\,dx,\quad n = 8, 16$

Compare and comment on the number of function evaluations needed for the various methods to obtain a given accuracy.

A14. For the following integrals, compute the trapezoid rule approximations $T(f;[a,b])$, $T(f;[a,\frac{a+b}{2}])$ and $T(f;\frac{a+b}{2},b])$, and verify the estimate given in the adaptive trapezoid error estimate.

a. $\displaystyle\int_0^{\sqrt{\pi/2}} x\sin(x^2)\,dx.$ **b.** $\displaystyle\int_{1/3\pi}^{1/\pi} \sin(1/x)\,dx.$

Use the adaptive trapezoid rule to find approximations to within 10^{-2} for the integrals.

Answer

a. With $a = 0$, $b = \sqrt{\pi/2} = 1.2533$, and $c = \frac{a+b}{2} = \sqrt{\pi/8} = 0.6267$, find $f(a) = 0$, $f(b) = 1.2533$, $f(c) = 0.2398$.
Then $T(f,[a,b]) = (b-a)(f(a)+f(b))/2 = 1.2533\times(0+1.2533)/2 = 0.7854$, $T(f,[a,c]) = 0.0751$ and $T(f;[c,b]) = 0.4678$, yielding $T_2(f;a,b) = T(f;[a,c,b]) = 0.0751 + 0.4678 = 0.5430$ and error estimate $E_T(f;[a,c,b]) \approx |T(f;[a,c,b]) - T(f;[a,b])|/3 = |0.5430 - 0.7854|/3 = 0.0808$.
Explicit formula gives $E_{T_2}(f;a,b) = \frac{1}{12}(b-a)\big(f(a) - 2f(\frac{a+b}{2}) + f(b)\big) = \frac{1}{12}\times 1.2533 \times |0 - 2\times 0.2398 + 1.2533| = \frac{1}{12}\times 1.2533\times 0.7737 = 0.0808$ as before.
Subdividing gives $T_2(f;a,c) = 0.0472$, $E_{T_2}(f;a,c) \approx 0.0093$, $T_2(f;c,b) = 0.4616$ and $E_{T_2}(f;c,b) \approx 0.0021$ for a total error estimate of $0.0114 > 10^{-2}$.
Subdividing $[a,c]$ gives $T_2(f;a,\frac{a+c}{2}) = 0.0030$, $E_{T_2}(f;a,\frac{a+c}{2}) = 0.0006$, $T_2(f;\frac{a+c}{2},c) = 0.0373$, $E_{T_2}(f;\frac{a+c}{2},c) = 0.0017$.
Integral $I \approx T_2(f;a,\frac{a+c}{2}) + T_2(f;\frac{a+c}{2},c) + T_2(f;c,b) = 0.0030 + 0.0373 + 0.4616 = 0.5019$ with error estimate $E \approx 0.0006 + 0.0017 + 0.0021 = 0.0044$.
Exact value is $\int_0^{\sqrt{\pi/2}} x\sin(x^2)dx = 1/2$, within 0.5019 ± 0.0044.

b. With $a = 0.1061$, $b = 0.3183$, need to subdivide $[a, b]$ at $c_0 = 0.2122$, then $[a, c_0]$ at $c_1 = 0.1592$, and $[a, c_1]$ at $c_2 = 0.1326$. Integral $I \approx T_2(f; a, c_2) + T_2(f; c_2, c_1) + T_2(f, c_1, c_0) + T_2(f, c_0, b) = 0.0178 + 0.0135 - 0.0340 - 0.0577 = -0.0604$, error estimate $E \approx E_{T_2}(f; a, c_2) + E_{T_2}(f; c_2, c_1) + E_{T_2}(f, c_1, c_0) + E_{T_2}(f, c_0, b) = 0.0017 + 0.0003 + 0.0025 + 0.0016 = 0.0061 < 10^{-2}$. Exact value $I = -0.063048$; within -0.0604 ± 0061; error 0.0026.

A15. Write a Matlab function to estimate $\int_a^b f(x)\,dx$ with accuracy ϵ using an adaptive version of the composite trapezoid rule. Test your code by estimating the integrals below to accuracy 10^{-4}.

a. $\displaystyle\int_0^{\sqrt{3\pi}} x\sin(x^2)\,dx$ **b.** $\displaystyle\int_{1/3\pi}^{4/\pi} \sin(1/x)\,dx.$

For **a**, the exact answer is 1. What is the minimum value of n for which the trapezoid rule yields an approximation to within 10^{-4}? How does this compare with the number of nodes required for the adaptive trapezoid rule?

E16. In a multivariable calculus and in statistics courses, it is shown that

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/2\sigma^2}\,dx = 1$$

for any positive σ. The function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/2\sigma^2}$$

is the *normal density function* with *mean* $\mu = 0$ and *standard deviation* σ. The probability that a randomly chosen value described by this distribution lies in $[a, b]$ is given by $\int_a^b f(x)dx$. Approximate to within 10^{-5} the probability that a randomly chosen value described by this distribution will lie in $[-k\sigma, k\sigma]$ for $k = 1, \dots, 5$.

E17. The study of light diffraction at a rectangular aperture involves the Fresnel integrals

$$c(t) = \int_0^t \cos(\tfrac{\pi}{2}w^2)\,dw \quad \text{and} \quad s(t) = \int_0^t \sin(\tfrac{\pi}{2}w^2)\,dw\,.$$

Construct a table of values for $c(t)$ and $s(t)$ that is accurate to within 10^{-4} for values of $t = 0.1, 0.2, \dots, 1.0$.

ch